Lynde Creek

An Analysis of the Lynde Creek Watershed with ArcGIS Pro

Team Number: 8

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Abstract

The framing of the watershed perimeter through the process of watershed delineation offers an interesting perspective on the inner workings of its hydrological characteristics, drainage patterns, and influence on the surrounding areas. Previous examinations into watershed delineation have resulted in major progress in terms of the tracking of adverse environmental effects on the water bodies and the feasibility of installing stable infrastructure within and surrounding the watershed. These examinations commonly employ the use of digital elevation models (DEM) to optimize time and reduce the possibility of human error when mapping the topography of the watershed. Similarly, our report utilizes the DEM system within the software, ArcGIS Pro, to define the exact perimeter of the Lynde Creek watershed. Through our research and application of the program, we found that our measure of the perimeter and area of the Lynde Creek watershed matches that of the report released by the Central Lake Ontario Conservation Authority, which is an organization that oversees a majority of reports conducted on the Lynde Creek watershed. Additionally, our identification of the slope of the watershed leading toward Lake Ontario matches the natural topography and drainage of the water system.

1.0 Introduction

The purpose of this report is to analyze the Lynde Creek watershed through the process of watershed delineation and the application of course concepts. This report will also describe all the necessary steps in the application of watershed delineation on the software ArcGIS Pro, as well as describe why they were included.

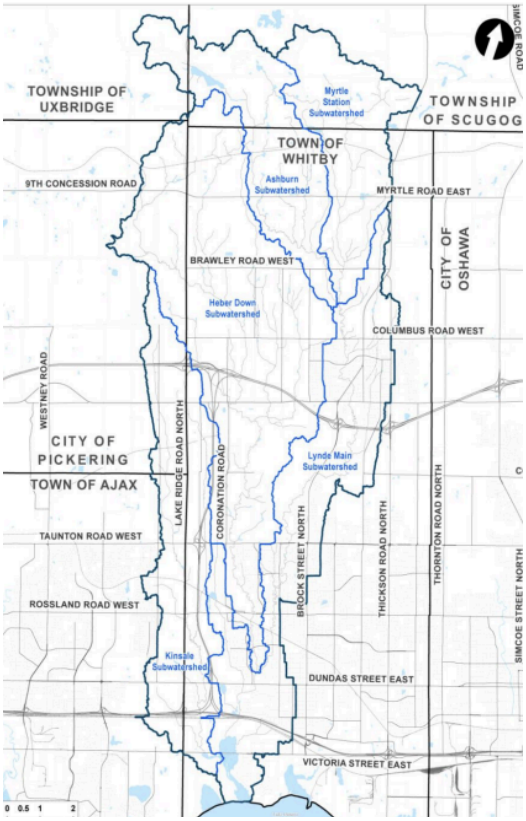

Figure 1. Map of Lynde Creek Watershed [4]

1.1 Lynde Creek

1.1.0 Location and Surrounding Area

The Lynde Creek watershed is divided into 5 subwatersheds: Lynde Main, Heber Down, Kinsale, Ashburn, and Myrtle Station, as seen in Figure 1. The mouth of the watershed opens south into Lake Ontario. The majority of the Lynde Creek watershed is located within the Town of Whitby,

with extensions into the adjacent towns and cities of Pickering, Ajax, Scugog, and Uxbridge. Due in part to the high levels of infrastructural and urban developments in the aforementioned municipalities since 2008, the watershed has undergone some noticeable shifts in structure and shape [1]. Developments in transportation and residential expansion, such as the addition of 17 new road crossings, including Highway 407 & 412, and the adjustment of the profile of Highway 401 [2], have reduced the amount of impervious surfaces, which increases the risk of flooding in specific areas [3].

1.1.1 Ecosystem

The Lynde Creek watershed contains two coastal wetlands along the shoreline of Lake Ontario, Cranberry Marsh and Lynde Creek Marsh [5]. The Cranberry Marsh and Lynde Creek ecosystems support high levels of biodiversity by providing vital habitats to a range of migratory birds, fish, amphibians, mammals, and submerged aquatic vegetation [6]. The biodiversity in both marshes is maintained due to the fluctuation in their water levels that results from breaches in their barrier beach systems. Both Lynde Creek Marsh and Cranberry Marsh are separated from Lake Ontario through a barrier beach system, which is crucial to shoreline protection and restricts water intake and output into Lake Ontario when there is high tide. However, the barrier beaches have been noticeably degrading over time, which has increased the number of breaches, rendering the wetlands more vulnerable to climate change, erosion, and diminished sand supply [7].

Figure 2. Map of Cranberry Marsh and Lynde Creek Marsh [5]

1.2 Importance of Watershed Declination

Within the fields of hydrology and water management, the analysis of watersheds and their characteristics is paramount to understanding ecological systems and framing engineering decisions. A key process within watershed analysis is watershed delineation, referring to the process of identifying the boundaries of the drainage areas within a watershed [8]. The data collected from watershed delineation can be utilized in numerous fields and professions, including infrastructure planning, flood and erosion control, climate planning, pollution management, and land development [8][10]. Additionally, the identification of the regions that the watershed's perimeter encompasses allows engineers to be aware of the multiple governmental jurisdictions that their project must abide by [9].

The misrepresentation of a watershed's perimeter may result in inaccurate tracking of the pollution distribution within the watershed, unstable infrastructure, and unpreparedness for adverse natural processes. A common cause of such misrepresentation is the use of a delineating watershed algorithm based on a digital elevation model (DEM), which is a program that eliminates the need to manually model the topography of the watershed by employing technology to monitor the continuous variation of relative elevation values [10]. One such

example stems from DEMs that leverage stereo viewing from optical satellites, in which the system possessed several observation flaws, including disruptions from sound waves and an inability to register tree heights [11].

2.0 Methodology

2.1 Lynde Creek Watershed Delineation

A watershed delineation for Lynde Creek was generated utilizing ArcGIS Pro. The analysis uses the Spatial Analyst extension, specifically the standard hydrological analysis tools. The purpose of the methodology was to follow an ordered workflow that converts topographic data into a hydrologically accurate watershed boundary. Figure 3 shows the summarized linear work flow of this section.

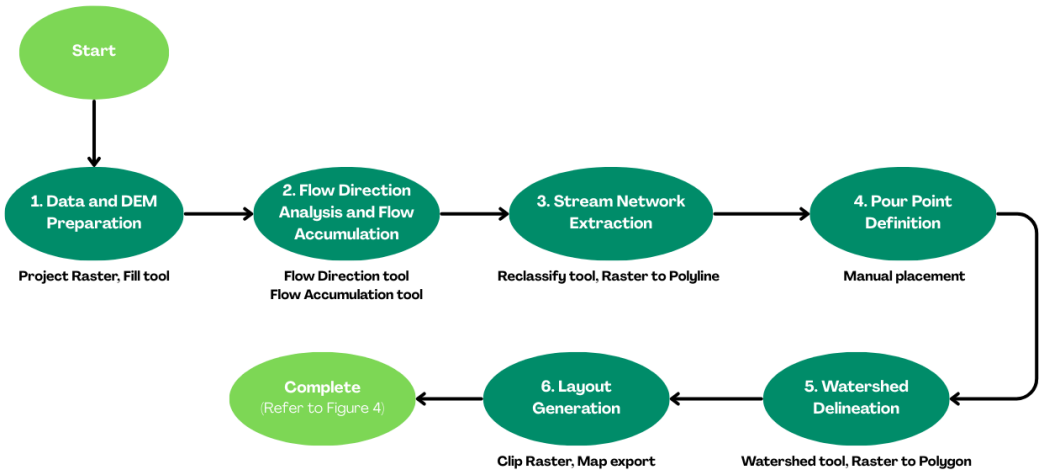

Figure 3. Flow chart of the Main steps in the watershed delineation process.

2.2 Data Preparation and Digital Elevation Model (DEM) Preparation

A Digital Elevation Model (DEM) was used for data references of our Lynde Creek analysis. The DEM was put into a coordinate system called WGS 1984 UTM Zone 17N to ensure distances and areas are correct. This step is important because DEM tools need to have the coordinates to identify waterflows and directions for an effective hydrological analysis.

To ensure our data was prepared, sinks in the DEM need to be removed by filling them in. These low spots need to be removed as they can disturb the water flows on the surface. Thus, the Fill tool was used, which is found under Spatial Analyst Tools and Hydrology on our projected DEM map. This is a critical step as it helps remove dips caused by problems with the data or mistakes when we were filling in the gaps. The goal is to make sure that the water can keep flowing without any interruptions all the way across the land.

2.3 Flow Direction Analysis and Flow Accumulation

At this time, the filled DEM is prepared to identify where water flows. We used the Flow Direction tool to do this. This tool uses the D8 algorithm to determine which way water moves from each cell to one of the cells next to it. There are eight cells next to each one. This step is important because it shows us where water will run off the surface. The DEM and the Flow Direction tool are necessary for understanding how water moves in the area because we need to find the direction of surface runoff to do hydrological analyses with the DEM.

So we used the Flow Accumulation tool with the flow direction raster. The Flow Accumulation tool counts how many cells are draining into each cell. This helps us find areas where water is really flowing together. When we look at the results, the cells with high flow accumulation values are actually the stream channels and the paths that water follows to drain. The Flow Accumulation tool is effective at showing us these stream channels and drainage pathways.

2.4 Stream Network Extraction

To find the stream network, we used the Reclassify tool on the flow accumulation raster. We chose a threshold value to separate the stream cells from the stream cells. This gave us a raster that we then changed to a vector format using the Raster to Polyline tool. This tool helped us create a stream network that shows Lynde Creek and the smaller streams that flow into it which are the tributaries of Lynde Creek.

2.5 Pour Point Definition

To create a watershed boundary, a special kind of map feature called a pour point was made. Then a point was put by hand at the mouth of Lynde Creek. This was done to make sure it lined

up perfectly with the streams that were found. It is very important to place the pour point accurately at the mouth of Lynde Creek because the pour point is where the water flows out of the area of land that drains water. Thus, the pour point directly affects how the rest of the land is divided.

2.6 Watershed Delineation

The Watershed tool was applied using the flow direction raster and the pour point feature to delineate the upstream contributing area. The resulting watershed raster was converted to a polygon using the Raster to Polygon tool to enable area calculation and cartographic presentation. The final polygon represents the delineated Lynde Creek watershed.

2.7 Layout Generation

Following this, we generated a layout formatting from ArcGIS to prepare for exporting the data. We organized the map with the visually displayed information/watershed and organized the data with labels and titles. Finally, we had a complete, accurate and complex watershed map to show the delineation.

3.0 Results

3.1 Delineated Watershed Boundary

The ArcGIS workflow described in section 2.0 produced the delineated watershed boundary map shown below in Figure 4. The DEM region downloaded from the OpenTopography database was selected to cover an area that extended beyond the anticipated Lynde Creek watershed boundary. Doing so ensured that all five of Lynde Creek's subwatersheds were fully captured in the watershed delineation process and that no contributing drainage areas were excluded. This is reflected in Figure 5, which depicts the mapped Lynde Creek watershed boundary having an area of 50.68 mi^2 [131.26 km^2] and a perimeter of 238,068.11 ft. (72.56 km). These values are highly consistent with those reported by the Central Lake Ontario Conservation Authority, which lists the watershed area at approximately 130 km^2, supporting the accuracy and reliability of the watershed delineation process conducted in ArcGIS Pro for Lynde Creek.

3.2 Observations

The watershed boundary shown in Figure # has a significantly larger longitudinal extent in the northwest direction from the outlet at Lake Ontario than its transverse width in the northeast-southwest direction. Due to the flow converging at this outlet to Lake Ontario to leave the watershed, the Lynde Creek watershed can be classified as an open watershed. Elevations in the watershed vary considerably, ranging from a low of 74.842 m near the outlet (orange) to a high of 408.318 m in the rivers' headwater region (blue). As observed by the colour gradient going from blue and green in the north to yellow and orange in the south, the elevations gradually decrease towards Lake Ontario. This pattern is consistent with the natural flow direction of runoff, which is downslope. In this watershed, water flows downhill through the streams depicted in Figure 4 to discharge to Lake Ontario.

A summary of Figure 4's geometric characteristics is listed in Table 1 below.

Table 1. Lynde Creek Data from Delineated Watershed Boundary Map

Measurements/Data	**Values**	**Units**
Maximum elevation	408.318	m
Minimum elevation	74.842	
Area	131.26	km^2
Perimeter	72.56	km

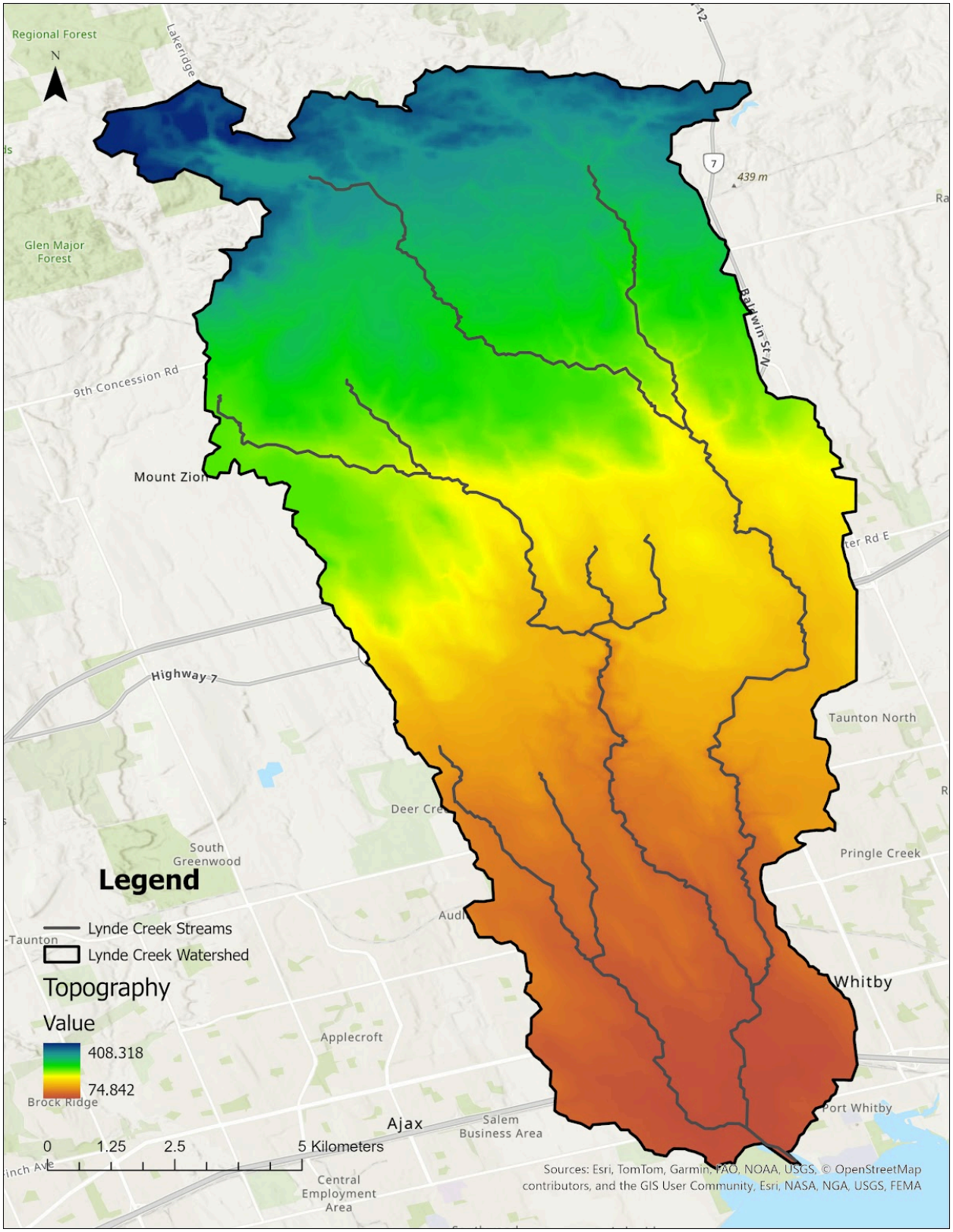

Figure 4. Lynde Creek Delineated Watershed Boundary Map [12]

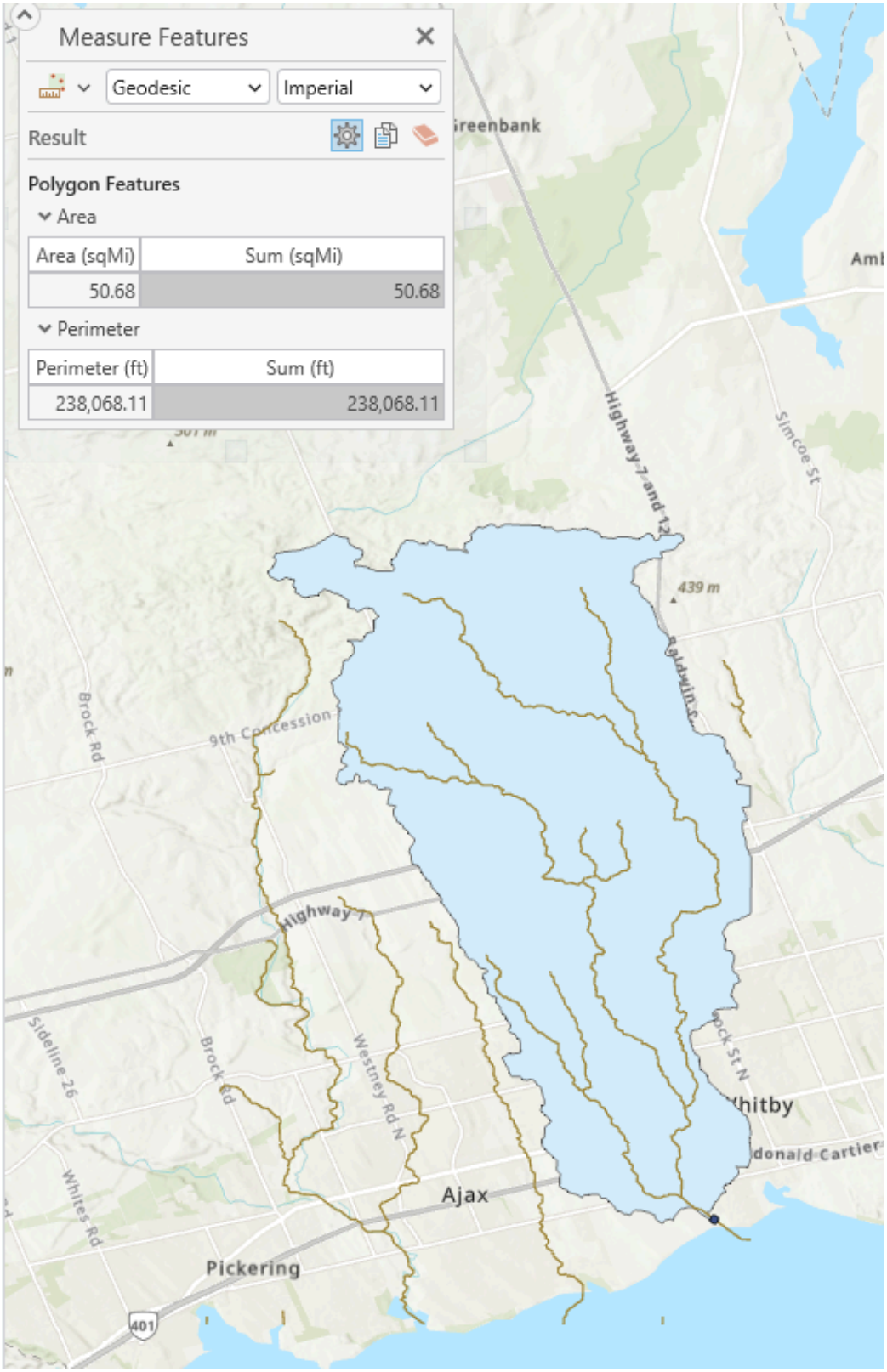

Figure 5. Computed Area and Perimeter of the Watershed Boundary on ArcGIS Pro [12]

4.0 Discussions

This section outlines the implications of our results for understanding the structure of the Lynde Creek watershed, as well as how these results can influence discussions surrounding the watershed and its development. Additionally, this section will also discuss the challenges we faced during the creation of this project and the application of the ArcGIS Pro software.

The results of the watershed delineation provide a deeper understanding of how water flows throughout the Lynde Creek watershed and its impact on the surrounding environment. The area of the watershed is approximately 131.26 km^2, which implies that excess water input into the watershed, such as precipitation and snow, will be spread around a larger area and thus will not cause a large amount of damage to the surrounding environment. The position of the Lynde Creek watershed along the shoreline of Lake Ontario also provides some information on the characteristics of the watershed and the influence that the lake holds over its water quality and the sustainability of its biodiversity. Due to the large size of the mouth of the Lynde Creek watershed, Lake Ontario holds heavy influence over the marshes within the watershed, such as Cranberry Marsh and Lynde Creek Marsh, notably during high tide. The natural fluctuation of water levels supports the ecosystems within the marshes. If exposed to high tide for prolonged periods of time, the species that rely on the surrounding wetlands are placed at risk of extinction.

Through this project, our group faced a number of different challenges in regards to adjusting As well as choosing an accurate pour point by accurately placing it and aligning with the stream network.

In comparison to existing literature, the Lynde Creek Watershed Plan found in the ArcGIS website, values like the area of the watershed are not exact. This could be a cause of a different DEM used, which may have caused discrepancies and differences in our data. This impacts the flow accumulation, and stream network extraction. However, the values are close enough, thus proving that our methodology was efficient in creating the delineated watershed boundary for Lynde Creek.

Citations

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